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Research Article

Structures and reactivity of monomeric MoO_x moieties supported on ZrO_2 (111) slab: A DFT study

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ABSTRACT

Supported molybdenum oxides have been widely employed as catalysts for variety of reactions including upgrading of bio-oil via hydrodeoxygenation. Herein, we perform spin-polarized density functional theory calculations to investigate the structures and reactivity of monomeric MoO_x species supported on ZrO_2 in the absence and presence of hydroxyl(s). Monomeric molybdenum moieties were found to have a mono-oxo (one molybdenyl), di-oxo (two molybdenyls) or a pero-oxo (one molybdenyl and Mo coordinated to a peroxide) configuration. Based on the formation energies, we find that mono-oxo moieties are the most stable among the three. This is also true for the moieties in the presence of hydroxyls on the support. Further, the computed vibrational frequencies of a mono-oxo moiety on a hydroxylated support is a closer match to the peaks observed in experiments. A thermodynamic analysis of the molybdenum oxide moieties on a hydroxylated support indicated that the dehydroxylation is favoured at high temperatures and low partial pressures of water. On adsorption of hydrogen on the oxygen of the support, hydrogen loses its electron and reduces Mo of the monomeric moiety without making an oxygen vacancy. We studied the activity of this reduced site for deoxygenation of anisole to benzene. The overall barriers for this reaction are comparable to previously computed barriers for deoxygenation of bio-oil model compounds on an oxygen-vacant molybdenum oxide surfaces, suggesting that this may be an important mechanistic pathway during a hydro-deoxygenation process.

1. Introduction

Bio-oil, produced as an intermediate after biomass pyrolysis, does not find direct utilization as a conventional transportation fuel because of its high oxygen content [1,2]. Therefore, bio-oil upgrading becomes necessary before its direct application as a motor fuel. Hydrodeoxygenation (HDO) is one of the key routes for upgrading bio-oil to biofuels [3,4]. Supported molybdenum oxide (MoO_x) have shown promise as a HDO catalyst under low pressures of H_2 [3,5–8]. In addition to HDO, supported molybdenum oxide find applications in several other reactions such as dehydrogenation [9], hydro-desulphurization, [10, 11] oxidative dehydrogenation of light alkanes, [9] selective oxidation of alcohols and low weight hydrocarbons [12–14].

Hydrodeoxygenation of model bio-oil compounds, such as m-cresol and anisole, has been studied on MoO_x supported on SiO_2, γAl_2O_3, TiO_2, ZrO_2, and CeO_2; [5,15] and it has been shown that the support plays an important role for this reaction. In this work, we use density functional theory calculations to probe the structures and reactivity of MoO_x supported on ZrO_2. We chose to study ZrO_2 as support because of several reasons. First, MoO_x supported on ZrO_2 (and TiO_2) showed the highest yields for hydrocarbons, while those supported on SiO_2 and γ-Al_2O_3 showed the lowest [5,15–17]. Second, Zr^{4+} is much less of an acid compared to Mo^{6+} and, unlike Ti^{4+}, [18,19] will not compete with Mo for the electrons donated from adsorbates, such as hydrogen, which are formed by dissociation of a H_2 molecule. The presence of an adsorbate may alter the nature of active site and its reactivity as discussed below. Lastly, although computational investigations have been preformed to understand the structures of MoO_x on TiO_2, Al_2O_3 and SiO_2 [20–31]; to the best of our efforts, we could not find any computational study probing the structures and reactivities of MoO_x supported on ZrO_2.

Several experimental studies carried out in the past have extensively investigated the structures and stability of supported molybdenum oxide catalysts [32–40]. The structure of supported molybdenum oxide moieties depend on several factors. These factors include, but are not limited to, the nature of the support, conditions at which these catalysts

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are prepared, surface coverage etc [32,35,41–44]. The prepared catalyst consists of monomeric, dimeric or polymeric molybdenum oxide moieties; all of which may be active for bio-oil up-gradation reaction. At low loadings (< 15% monolayer coverage), it was shown previously [45] that the supported ZrO_2 catalyst is dominated by the mono-molybdate species, which is the focus of the current work. Despite, significant experimental efforts probing the structures of supported molybdenum oxides, computational investigations are limited to a handful of studies [20–31]. It still remains unclear whether the surface MoO_x moieties exist as mono-oxo or a di-oxo or pero-oxo structures. The coordination number of Mo in these moiety species is also a subject of debate. For example, both penta-coordinated and tetra-coordinated Mo have been predicted [37,46–48]. Therefore, one of the objectives of this work is to investigate the monomeric moieties of MoO_x supported on ZrO_2 under dehydrated conditions. Hydro-deoxygenation with supported MoO_x are done at elevated temperatures, where a dehydrated surface forms an appropriate model for comparison. For dehydrated conditions, we assume no adsorbed water is present on the surface, however, surface hydroxyls may be present. A real experimental catalyst will have hydroxyls on the surface, and therefore, their affect on the structures of molybdenum oxide moieties is an interesting question to investigate.

We also study reactivity of MoO_x moieties in presence of a hydrogen adsorbate. There are several reasons to study adsorption of a hydrogen atom on the oxygen of an oxide. For example, binding energy of hydrogen forms a descriptor for breaking of a C–H bond [49,50]. The activation energy is lower when the binding energy of H is stronger, which is a correlation from the Brønsted–Evans–Polanyi relationship [51]. There are several previous reports which have shown that different species can exchange electrons when present on the same support [52–60]. Hydrogen is a strong base and on adsorption on to the support, it may reduce metal ions, affecting its catalytic activity. In this work, we are interested in studying the adsorption of hydrogen to understand the effect of hydroxyls on the structures of monomeric MoO_x moieties. We are also interested in understanding the role, if any, of hydrogen in activating the MoO_x moieties during hydro-deoxygenation reactions, which are carried in reducing atmospheres of hydrogen. For this purpose, we compute adsorption and dissociation of anisole (a bio-oil model compound) to benzene, which is the major product of anisole hydrodeoxygenation [16,17].

2. Methodology

2.1. Stoichiometric ZrO_2 (111) Slabs

The most stable polymorph of ZrO_2 has a monoclinic structure with the periodicity given by space group of $P2_1/c$. In this work, we study the (111) facet of ZrO_2 as it has been reported to have the lowest surface energy among the different low index planes [61,62]. The model slab of ZrO_2 having only (111) surface exposed [denoted as 'ZrO_2(111)'] was prepared by cleaving along the Zr–O bonds using a previously developed python routine [63]. The algorithm generates six terminations, each of which are optimized before computing surface energies. The simulation cell had a surface of 4 × 4 units. The slabs were four layer thick and a vacuum of 15 Å was used in our simulations. The front and top views of the optimized structure corresponding to each termination and their respective surface energy are given in Supplementary Information (see Figure S1). We note here that the optimization algorithms used in this work are not able to capture major surface reconstruction, if any. The computational expense involved in global optimization techniques limits its usage and are outside the view of the current work.

The surface energy of the ZrO_2 slab (E_{surf}) was obtained by the relation

$$E_{surf} = \frac{E_{slab} - (n \times E_{bulk})}{2A}. \quad (1)$$

Here $2A$ is the total surface area, E_{slab} is the total energy of the ZrO_2 slab and E_{bulk} is the total energy of the bulk ZrO_2 per formula unit of ZrO_2. Further, n represents the total number of ZrO_2 units in the model slab. The termination with the lowest surface energy (i.e., termination 1 in Figure S1 of Supplementary Information) was used for future calculations. The surface of ZrO_2(111) has one type of Zr ion and two different types of oxygen ions. One of the oxygen has a 2-fold coordination and the other has a 3-fold coordination. The Zr–O bond lengths varies between 2.01 and 2.33 Å. We find that the Bader charge on Zr to be 2.65e, which we assign as a formal charge of +4. A two-fold oxygen ion has a Bader charge of −1.25e and a 3-fold oxygen has a Bader charge of −1.35e.

2.2. Computational details

We used the VASP software bundle [64–67] to perform quantum mechanical calculations based on density functional theory (DFT) [68]. Spin-polarized wave functions were constructed using plane waves having kinetic energy less than 400 eV. For volume optimization, a higher value of 550 eV was used. The exchange–correlation functional was obtained from the Perdew–Burke–Ernzerhof formulation [69]. The core electrons were treated with the PAW potentials and the number of explicit electrons used in our calculations were: twelve for Zr and Mo, six for O, four for C, and one for H. Grimme's D3 method [70] was applied to compute dispersion corrections. The Brillouin zone had 2 × 2 × 1 k-points, which were enough to converge the energies to within 0.01 eV. To penalize the tendency of the DFT functional to delocalize the d-electrons, we used the PBE+U method. A U_{eff} value (where $U_{eff} = U - J$) of 2 eV and 3 eV, as suggested previously, [71,72] were added to the Mo and Zr $3d$ states, respectively. DFT with a hybrid functional is a satisfactory fix at a much larger computational cost—prohibitive for the size of systems studied in this work.

The electronic wave-functions were converged from an initial guess in an electronic loop with an energy convergence criteria of 10^{-5} eV. For frequency calculations, a criteria of 10^{-6} eV was used. Only the atoms forming the Mo moiety and the adsorbates (H or OH) were allowed to move during the frequency calculations. We also tested a couple of systems by unfixing the surface atoms and we found minor changes in the reported frequencies. Frequencies were scaled using a factor of 1.03 as suggested previously for the PBE functional [73]. Using this scaling factor for the gas-phase MoO_3 we get a closer match to the experimental frequencies. For example, we get a Mo=O stretching frequency of 938 cm^{-1} using PBE+U method, which after scaling becomes 967 cm^{-1} and matches closely with the experimentally reported value of 976 cm^{-1} [74]. The nuclear coordinates were relaxed till the forces on each ion were less than 0.02 eV/Å. The geometries were optimized with a stricter criteria of 0.01 eV/Åbefore performing frequency calculations. We used the Bader's method for partitioning the total electron density into individual ions [75,76]. The integrated electron density on a particular ion gave us the total electrons on that ion. The net charge was obtained by subtracting the explicit electrons from this number. Further, we integrated the difference in up-spin and down-spin densities on a particular ion to get the net-spin on that ion.

The activated complex connecting the reactant and the product for anisole dissociation was obtained using the climbing-image (CI) and the improved-tangent (IT) versions of the nudged elastic band (NEB) method [77–79]. The band connecting the stationary ends of the reaction was discretized using six to eight images—the starting guess of which was obtained using linear interpolation between the two ends. The NEB method puts a soft constraint on the distance between images using springs. We used a uniform spring constant of 5 eV/Å^2 for connecting all the images. The iteration loop in NEB was terminated when the maximum NEB forces on each ion was less than 0.05 eV/Å.

3. Results and discussion

3.1. Structures of monomeric MoO_x moieties on pristine ZrO_2 slab

We begin by studying the formation energies of monomeric MoO_x moieties supported on $ZrO_2(111)$. We created several permutations and orientations of monomeric moieties with formula MoO_x on a pristine $ZrO_2(111)$. The subscript 'x' denotes the number of oxygen atom added with Mo to the pristine $ZrO_2(111)$. We explored moieties with $x = 3$ and $x = 4$ in this work. After optimization, Mo also gets coordinated with the surface oxygen atoms of $ZrO_2(111)$. We do not use these oxygens for denoting the moieties below. We find MoO_x moieties assume three different configurations: (a) mono-oxo moiety having one molybdenyl (Mo=O bond), (b) di-oxo moiety having two molybdenyls and (c) pero-oxo moiety having one molybdenyl and Mo coordinated to oxygen forming a peroxide. Mono-oxo and di-oxo moieties were obtained by placing a MoO_3 on $ZrO_2(111)$ and a pero-oxo moiety is obtained by placing a MoO_4 on $ZrO_2(111)$. The optimized structures of the moieties in the most stable orientation are shown in Fig. 1. Several previous calculations have reported the possibility of pero-oxo species on supported metal oxides [19,80–82].

In the mono-oxo moiety, Mo is coordinated with four oxygens and the moiety assumes a distorted tetrahedral structure (see Fig. 1a). Mo forms a double bond with one of the oxygen atoms, as evident from the short Mo–O bond length of 1.69 Å, and interacts with the surface of $ZrO_2(111)$ through three bridging oxygens. We find that the bridging Mo–O bond length varies from 1.79 to 1.85 Å. In the di-oxo moiety, the central Mo ion forms two Mo=O bonds and binds to the ZrO_2 surface through two bridging oxygens (see Fig. 1b). In the pero-oxo moiety, Mo is coordinated to five oxygens and the moiety assumes a geometry of distorted square pyramidal. Mo ion forms a double bond with one of the oxygen ions and coordinates to four bridging oxygens—one of them forms a peroxide with the surface oxygen ion (see Fig. 1c). The bond lengths of Mo with the bridging oxygen varies from 1.8 to 2.20 Å.

We computed the formation energies (ΔE_f) of these moieties using:

$$\Delta E_f = E_{Mo_nO_m/ZrO_2} - E_{ZrO_2} - nE_{Mo} - \frac{m}{2}E_{O_2}. \quad (2)$$

Here $E_{Mo_nO_m/ZrO_2}$ is the energy of the MoO_x moiety supported on $ZrO_2(111)$, E_{ZrO_2} is the energy of pristine $ZrO_2(111)$, and E_{Mo} is the energy of the bulk molybdenum in BCC form computed per Mo atom. E_{O_2} is the energy of the isolated O_2 molecule. The formation energies for each moiety are given in Fig. 1. By comparing the formation energies of mono-oxo, di-oxo and pero-oxo moieties, we conclude that mono-oxo moiety is the most stable among the three. Because di-oxo and mono-oxo species have the same number of atoms, the one having more negative formation energy is the more stable species. The pero-oxo moiety, on the other, is enthalpically and entropically less favourable than mono-oxo moiety at 1 atm pressure as it is formed by a consumption of an additional $\frac{1}{2}O_2$.

The Bader charge on Mo in all the three moieties is between 2.5e−2.6e (see Table S1 in Supplementary Information) and the net-spin on Mo is 0. We assign Mo to be in the state having a formal charge of +6. Bader charge analysis also suggests that both mono-oxo and di-oxo moieties are present as $[MoO_4]^{2-}$ and pero-oxo moiety is present as $[O_4Mo(O_2)]^{2-}$.

We now compare the results of frequency calculations with the experimental findings. We summarize the results of our frequency analysis in Table 1 along with experimental Raman-active frequencies and assignments. Previous experiments have reported a broad band at 970 cm^{-1} and a diffuse band at 845 cm^{-1} in the Raman spectra for a ~13% monolayer coverage [45]. For such low coverages, the supported catalyst is dominated by the mono-molybdate species and forms a relevant comparison for our calculations.

We focus on qualitative findings in this work and do not expect a quantitative match with the experimental findings because of several reasons. First, we have studied only (111) surface of ZrO_2. However, in experiments several surfaces of ZrO_2 may be present on which MoO_x may be deposited. Secondly, the experimental sample may have populations of all the three types of moieties. Thirdly, we computed a defect-free surface of ZrO_2, which is not valid for a physical sample. Fourthly, we have assumed no adsorbates (e.g. hydroxyls) are present on $ZrO_2(111)$—an assumption which may not be valid under experimental conditions. We study the effect of hydroxyls in the next section.

As shown in Table 1, the peak positions observed in experiments are a closer match to frequencies obtained for a mono-oxo moiety. For example, the frequency obtained at 974 cm^{-1} matches with the peak position at 970 cm^{-1} and the frequency at 870 cm^{-1} is a close match to the peak position at 845 cm^{-1}. We show below that 870 cm^{-1} red-shifts to 849 cm^{-1} in the presence of hydroxyl on the surface—rendering our calculations a much closer match with the experimental results. We, however, differ from literature on the peak assignments. The peak at 970 cm^{-1} was assigned to $(O=)_2Mo$ symmetric stretching of the di-oxo moiety, [45] whereas, we find that the frequency at 974 cm^{-1} corresponds to Mo=O stretching of the mono-oxo moiety. Similarly, the peak at 845 cm^{-1} was assigned to Mo–O–Zr vibration, [45] whereas, we find that the frequency at 870 cm^{-1} corresponds to O–Mo–O vibration. The Mo–O–Zr vibrations, according to our calculations, appear below 700 cm^{-1}; for mono-oxo moiety, Mo–O–Zr vibrations appear at 680 and 668 cm^{-1}, and for the di-oxo species Mo–O–Zr vibrations are observed at 694 cm^{-1}. Therefore, we conclude that the experimental assignments requires to be revisited. We also note that the broad peak observed at 970 cm^{-1} and the diffuse peak observed at 845 cm^{-1} encompasses the vibrations of all three moieties.

Although, our work focusses on monomeric moieties, we also present frequency analysis of the most stable dimeric molybdate moiety; the molecular structure is given in Supplementary Information, Figure S2. The findings also gives credence to the analysis presented for the monomeric moieties. We summarize the results of our frequency calculations in the Supplementary Information (Table S1) along with experimental Raman-active frequencies at ~50% monolayer coverage. At these loadings, we assume dimeric moieties dominate. The frequency of the moiety obtained at 982 cm^{-1} matches well with the peak observed at 985 cm^{-1} in experiments and the frequency of 855 cm^{-1} is a close match to the peak observed at 837 cm^{-1} [39]. We also find a frequency of 973 cm^{-1}, which corresponds to the asymmetric Mo=O stretching mode of the dimeric moiety and expected it to be Raman inactive. It is observed in experiments that the peak at 970 cm^{-1} for low loadings of molybdenum oxide blue-shifts at higher loadings. This is consistent with our calculations. Note that it is possible a dimeric moiety may dissociate to form two monomeric moieties, which are more stable as shown previously for VO_x supported on TiO_2 [19]. We do not study these cases here.

3.2. Structures of monomeric $MoO_x/ZrO_2(111)$ in the presence of Hydroxyl(s)

We showed in the last section that mono-oxo $MoO_3/ZrO_2(111)$ is the most stable moiety under strictly dehydrated conditions (no hydroxyls on the surface). We, therefore, use this moiety as the baseline for performing future calculations. Hydroxyl(s) can be created by adsorbing hydrogen atom(s) on the $MoO_3/ZrO_2(111)$. We have studied various possibilities, which we outline below.

3.2.1. Adsorption of a Hydrogen atom

We begin by studying the adsorption of a hydrogen atom on the mono-oxo $MoO_3/ZrO_2(111)$. A hydrogen can be placed on the oxygen of the $ZrO_2(111)$ or on the oxygen of the MoO_3 moiety or on the Mo of the MoO_3 moiety. We have studied several possibilities and the most stable configuration is shown in Fig. 2a. Figs. 2b and 2c will be discussed below. In this configuration, the hydrogen adsorbs on

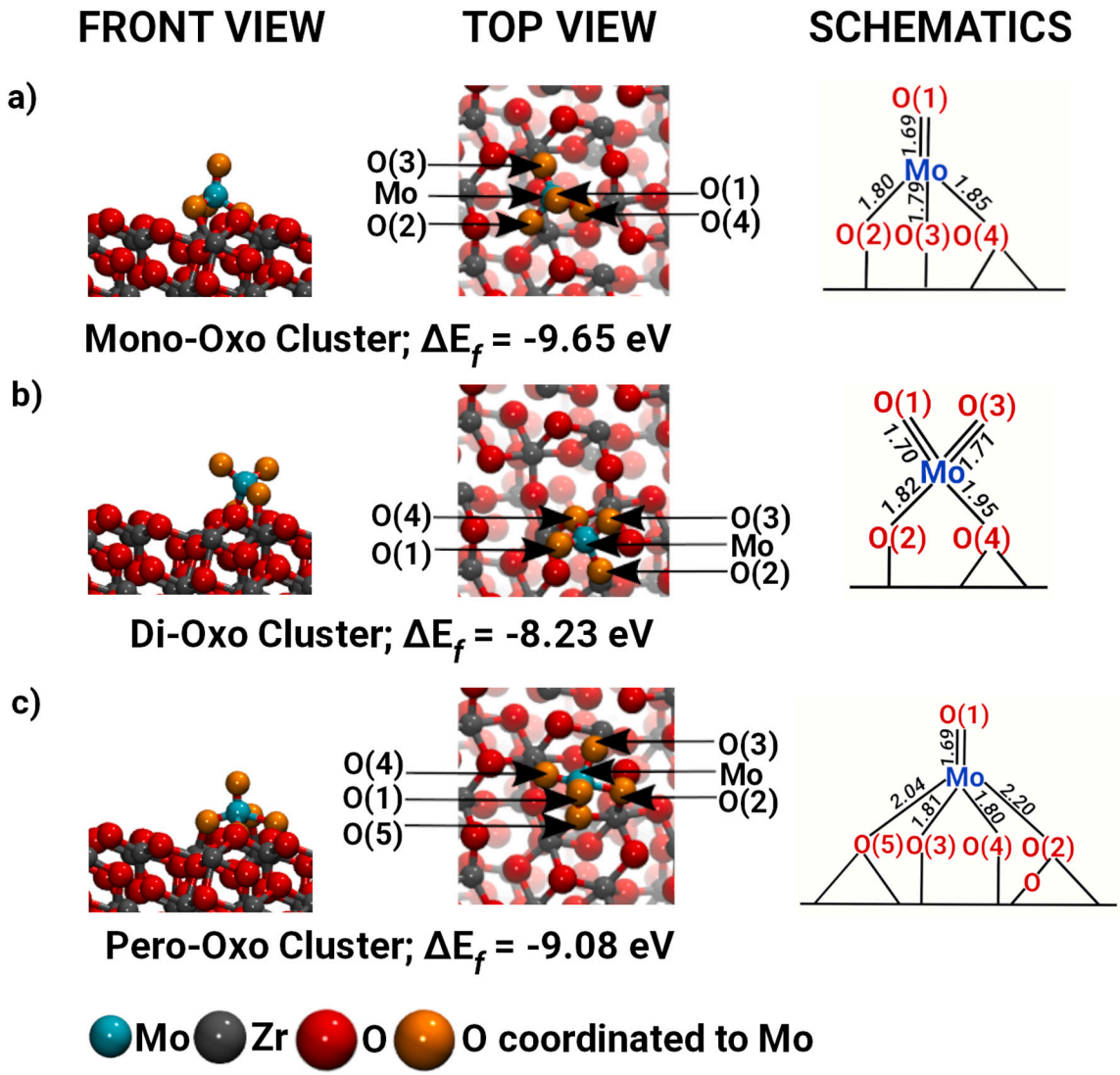

Fig. 1. Orthographic projections of MoO_x moiety supported on $ZrO_2(111)$. **(a)** Mono-oxo, **(b)** di-oxo, and **(c)** pero-oxo moiety. The formation energies computed using PBE functional are also shown. The right panel shows the schematic of the moieties along with Mo–O distances.

Table 1
Vibrational frequencies computed using PBE+U of strictly dehydrated molybdenum oxide moieties supported on $ZrO_2(111)$. Frequencies in the parenthesis are corresponding to unfixed surface atoms. The reported frequencies are scaled by a factor of 1.03.

Experimental[45]		Mono−oxo		Di−oxo		Pero−oxo	
Raman Freq. (cm^{-1})	Assigned	Freq. (cm^{-1})	Modes	Freq. (cm^{-1})	Modes	Freq. (cm^{-1})	Modes
970	$(O{=})_2MoO_2$ (symmetric)	974 (974)	Mo=O (stretching)	955 (956)	O=Mo=O (symmetric)	985	Mo=O (stretching)
845	Mo–O–Zr	870 (872)	O–Mo–O	941 (941)	O=Mo=O (asymmetric)	810	O–Mo–O
		801 (810)	O–Mo–O	782 (794)	O–Mo–O	848	Mo–O–O
		773 (785)				733	O–Mo–O

the neighbouring oxygen of the moiety and forms a hydrogen bond with the bridging oxygen of the moiety. We also show several selected configurations of adsorbed hydrogen on mono-oxo $MoO_3/ZrO_2(111)$ in Supplementary Information (Figure S3).

We find that the adsorption of hydrogen with respect to gas-phase $\frac{1}{2}H_2$ is energetically favourable, i.e., the energy change of reaction (ΔE_H)

$$MoO_3/ZrO_2(111) + \frac{1}{2}H_2 \longrightarrow MoO_3(H)/ZrO_2(111) \quad (3)$$

is negative (−0.34 eV). Hydrogen is a strong base and loses its electron on adsorption, which localizes on Mo ion of mono-oxo $MoO_3/ZrO_2(111)$. We plot the PDOS of mono-oxo $MoO_3/ZrO_2(111)$ before and after hydrogen adsorption in Figs. 3a and 3b, respectively. Fig. 3c will be discussed below. A plot of the iso-surface of the polaron (localized electron) electron density is also shown in the right panel in Fig. 3b. On adsorption, hydrogen donates its electron to the system forming a polaron on the Mo ion, which changes its state from a formal charge of +6 to +5. The net spin on Mo changes from 0 to 0.85 and the Bader

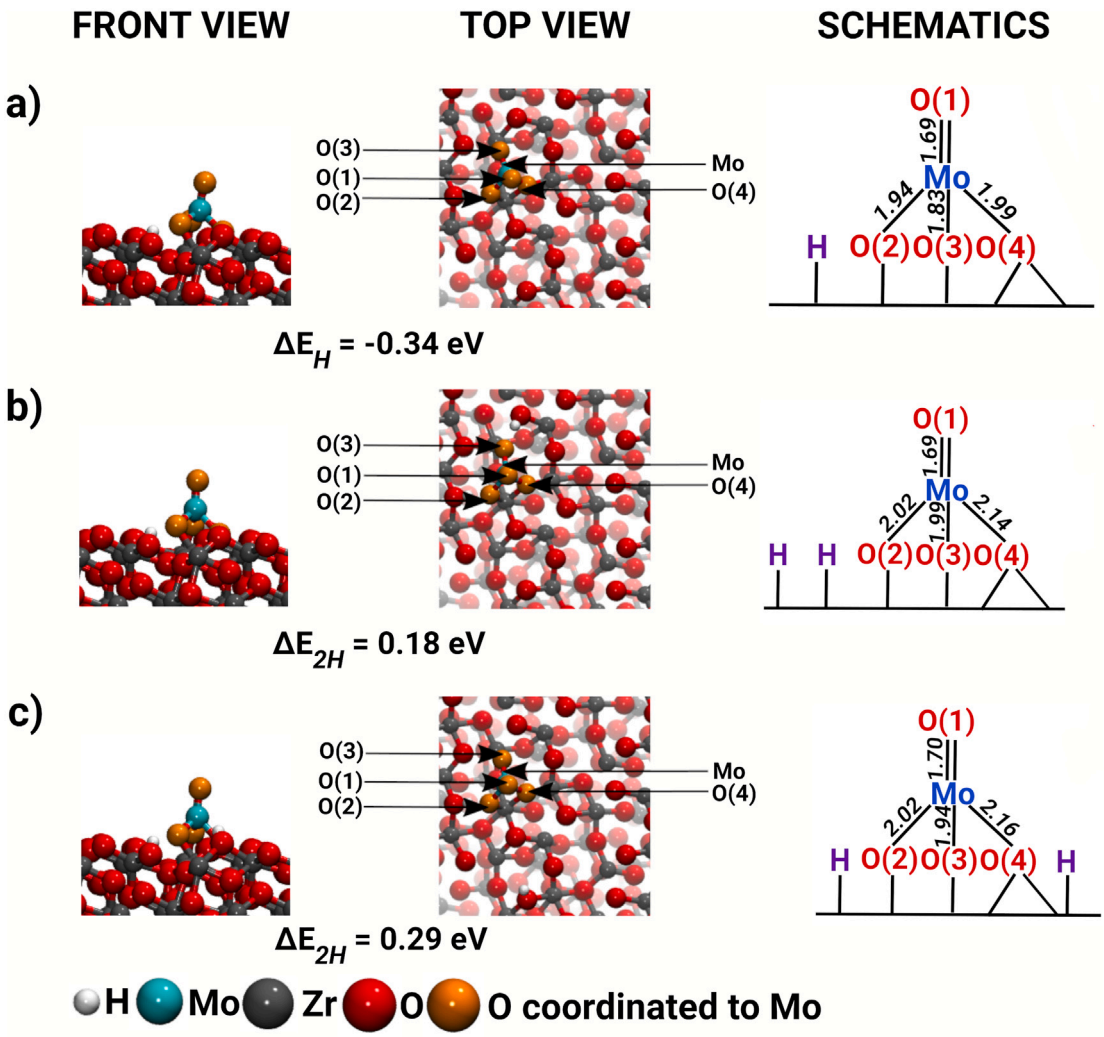

Fig. 2. Configuration of adsorbed hydrogen(s) on supported mono-oxo molybdenum oxide moiety in the most energetically favoured state. **(a)** Single H adsorbed, **(b)** two H adsorbed, and **(c)** two H adsorbed in a slightly less energetically favoured state. On reacting with O, structure (b) gave us the di-oxo moiety and structure (c) gave us the mono-oxo moiety. ΔE_{H} is adsorption energy of H atom with respect to $1/2H_2$ and $\Delta E_{2\mathrm{H}}$ is the adsorption energy of two H atoms with respect to H_2 molecule. The right panel shows the schematic of the moieties along with Mo–O distances.

charge reduces from 2.55 to 2.29. It was shown previously [83] that a hydroxylated ZrO_2 surface can be reduced. We tried to localize the polaron on a couple of neighbouring Zr ions but in both the cases, after optimization, the polaron was formed on the Mo ion. The permutations of polaron formation on Zr ion would be extremely large [54] and out of scope of the current work. As a result of polaron formation, Mo increases its bond length with the bridging oxygens (see Fig. 2a for details on bond lengths). The unpaired electron localizes in the *d*-orbital of the Mo ion and the filled state appears in the band gap of mono-oxo $MoO_3/ZrO_2(111)$ in PDOS suggesting that it will act as a strong base.

We also computed the frequencies of the mono-oxo $MoO_3(H)/ZrO_2(111)$. In general, we find there is a red-shift in the frequencies associated with the Mo atom due to the formation of a polaron on Mo. For example, Mo=O stretching frequency red-shifts from 974 to 958 cm^{-1} and O–Mo–O vibration red-shifts from 870 cm^{-1} to 783 cm^{-1}.

We conclude that the presence of a strong base like H atoms will influence the structures and reactivity of the supported molybdenum oxide moiety. This finding is in line with previous suggestion for vanadium-oxide moieties supported on TiO_2 [19].

3.2.2. Adsorption of two Hydrogen atoms

Beginning with the most stable mono-oxo $MoO_3(H)/ZrO_2(111)$, we studied the energetics and structures of placing the second hydrogen. The most stable configuration is shown in Fig. 2b, which is used for future studies. We also study a similar energy configuration shown in Fig. 2c for future exploration because this configuration gave us the most stable geometry after interacting with O, which is a strong acid. We also show several configurations of mono-oxo $MoO_3(H)(H)/ZrO_2(111)$ in Supplementary Information (Figure S4).

We find that the adsorption of second hydrogen is endothermic, i.e., the energy change of the reaction ($\Delta E_{2\mathrm{H}}$)

$$MoO_3/ZrO_2(111) + H_2 \longrightarrow MoO_3(H)(H)/ZrO_2(111) \quad (4)$$

is 0.18 eV. In comparison, the reaction where the moiety reacts with H_2 to produce H_2O and an oxygen vacancy requires an energy of 1.64 eV. The dissociation of hydrogen is also entropically favourable and will drive ΔG of the reaction to be negative at high temperatures. The second hydrogen prefers to sit in the neighbouring oxygen of the moiety forming a strong hydrogen bond with the oxygen of the MoO_3 moiety. Hydrogen on adsorption donates its electron, which forms another polaron on the Mo ion of the MoO_3 moiety. We find that the net spin on Mo is 1.73 and the Bader charge reduces to 2.01, which we attribute

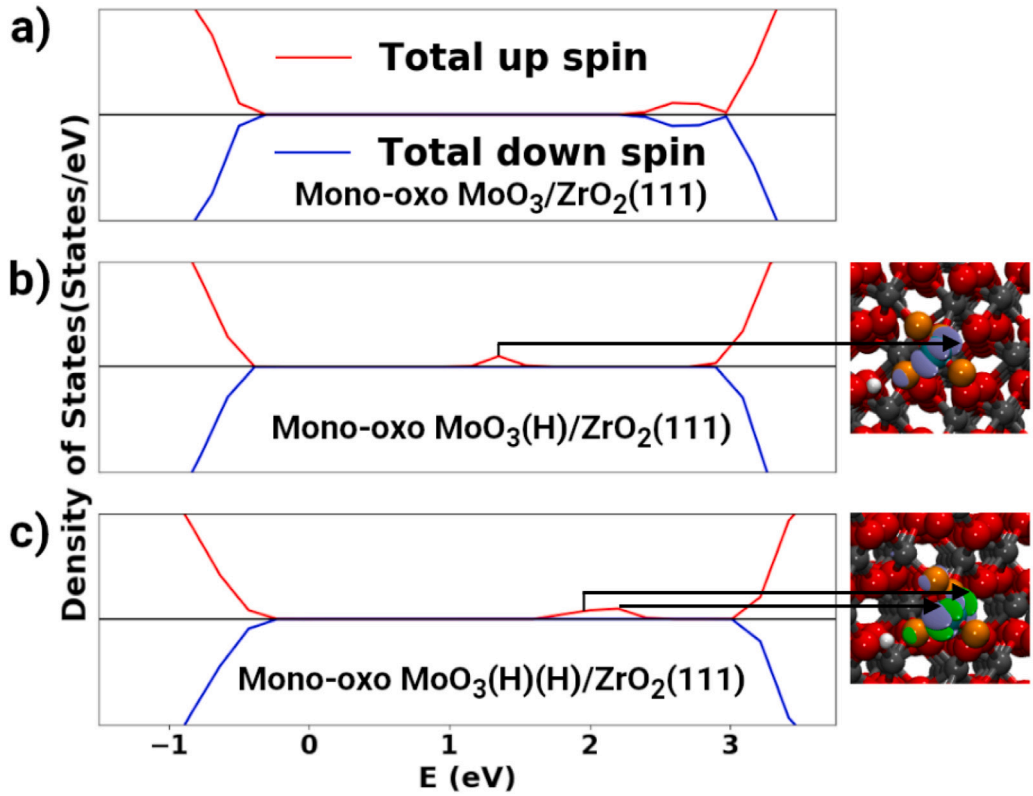

Fig. 3. Projected density of states (PDOS) of mono-oxo $MoO_3/ZrO_2(111)$ in the: **(a)** absence of H adsorbate, **(b)** presence of a single H adsorbate and **(c)** presence of two H adsorbates. For (b) and (c), we also show partial charge densities of the localized electrons for an iso-value of 0.03 e/Å^3 in green and blue colours. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to Mo having a formal charge of +4. The polaron appears in the band gap as shown in PDOS plot (Fig. 3c). Frequency analysis of the reduced moiety shows that Mo=O stretching frequency red-shifts to 935 and O–Mo–O vibration red-shifts to 710 cm^{-1}.

3.2.3. Reaction of O with $MoO_3(H)(H)$ moieties

In the previous section, we showed that placing hydrogen(s) on the mono-oxo $MoO_3/ZrO_2(111)$ creates polaron(s) on the Mo ion of the moiety. Experimentally, the catalyst is prepared by calcination in air, and therefore, we study the oxidation of the $MoO_3(H)(H)$ moieties with an O atom. We have considered following possibilities in our work. First, reacting the oxygen atom with the molybdenum oxide moiety, which would result in either a mono-oxo moiety or a di-oxo moiety (denoted as '$MoO_3(H)(H)(O)/ZrO_2(111)$'). Second, the oxygen can combine with the hydrogen on the surface to form OH coordinated to the Zr on the surface (denoted as '$MoO_3(H)(OH)/ZrO_2(111)$'). Third, the oxygen adsorbs as adatom on the surface. We found that the structure of oxygen adatom on mono-oxo $MoO_3(H)(H)/ZrO_2(111)$ was less stable, and therefore, not discussed here.

The structures of mono-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$, di-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$ and mono-oxo $MoO_3(H)(OH)/ZrO_2(111)$ are shown in Figs. 4a, 4b and 4c; respectively. All other less stable possibilities are shown in Supplementary Information (Figure S5). Note that on placing oxygen on the mono-oxo $MoO_3(H)(H)$ moiety, and after optimizing, Mo severed its interaction with one of the oxygen on the ZrO_2 surface. For forming mono-oxo moiety, it formed a Mo–O–Zr both with the added oxygen, whereas for forming the di-oxo moiety it formed a double bond with the added oxygen. In both the cases, we find the moieties are present as $[MoO_4]^{2-}$.

In comparison to strictly dehydrated MoO_3 moieties, there are slight changes in the bond lengths. Bader charge analysis and net spin indicates that Mo is in +6 state in all the moieties. We give the frequencies of the moieties in Table 2. The frequencies of mono-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$ are a closer match with experiments compared to strictly dehydrated mono-oxo $MoO_3/ZrO_2(111)$. The H on the hydroxyl forms a hydrogen bond with the oxygen of the molybdenum oxide moiety, which is responsible for flattening of the O–Mo–O vibrational potential. We, therefore, observe a red-shift in the vibrational, frequency from 870 cm^{-1} to 845 cm^{-1}. We also find that the presence of a H and OH on the surface does affect the frequencies of the molybdenum oxide moiety. On adding two more hydroxyls on the surface, we find negligible changes in the vibrational frequencies of the MoOx moieties, and therefore, the results are not reported here.

The energy (ΔE) of molybdenum oxide moiety with respect to mono-oxo $MoO_3/ZrO_2(111)$ and gas-phase water are also shown in Fig. 4. Interestingly, we find that the mono-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$ and the mono-oxo $MoO_3(H)(OH)/ZrO_2(111)$ have similar energies and, in comparison, the di-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$ is much less stable. However, the relative stability of di-oxo moiety increases in the presence of hydroxyls.

An O atom is a strong acid whereas the two polarons located on the Mo ion are strong bases. As a result, the oxygen atom withdraws the unpaired electrons, which were earlier localized on the Mo atom. The overall process of adsorbing two hydrogens and one oxygen consumes one water molecule and is equivalent to the following reaction:

$$MoO_3/ZrO_2(111) + H_2O \longrightarrow MoO_3(H)(H)(O)/ZrO_2(111) \text{ or } MoO_3(H)(OH)/ZrO_2(111) \quad (5)$$

This reaction creates hydroxyls on the surface and we study this reaction to understand the stability of $MoO_3(H)(H)(O)/ZrO_2(111)$ [or $MoO_3(H)(OH)/ZrO_2(111)$], in comparison to strictly dehydrated $MoO_3/ZrO_2(111)$, as a function of temperature and partial pressures of water. We computed the free energy (ΔG) of the reaction as:

$$\Delta G = \Delta H - T\Delta S \quad (6)$$

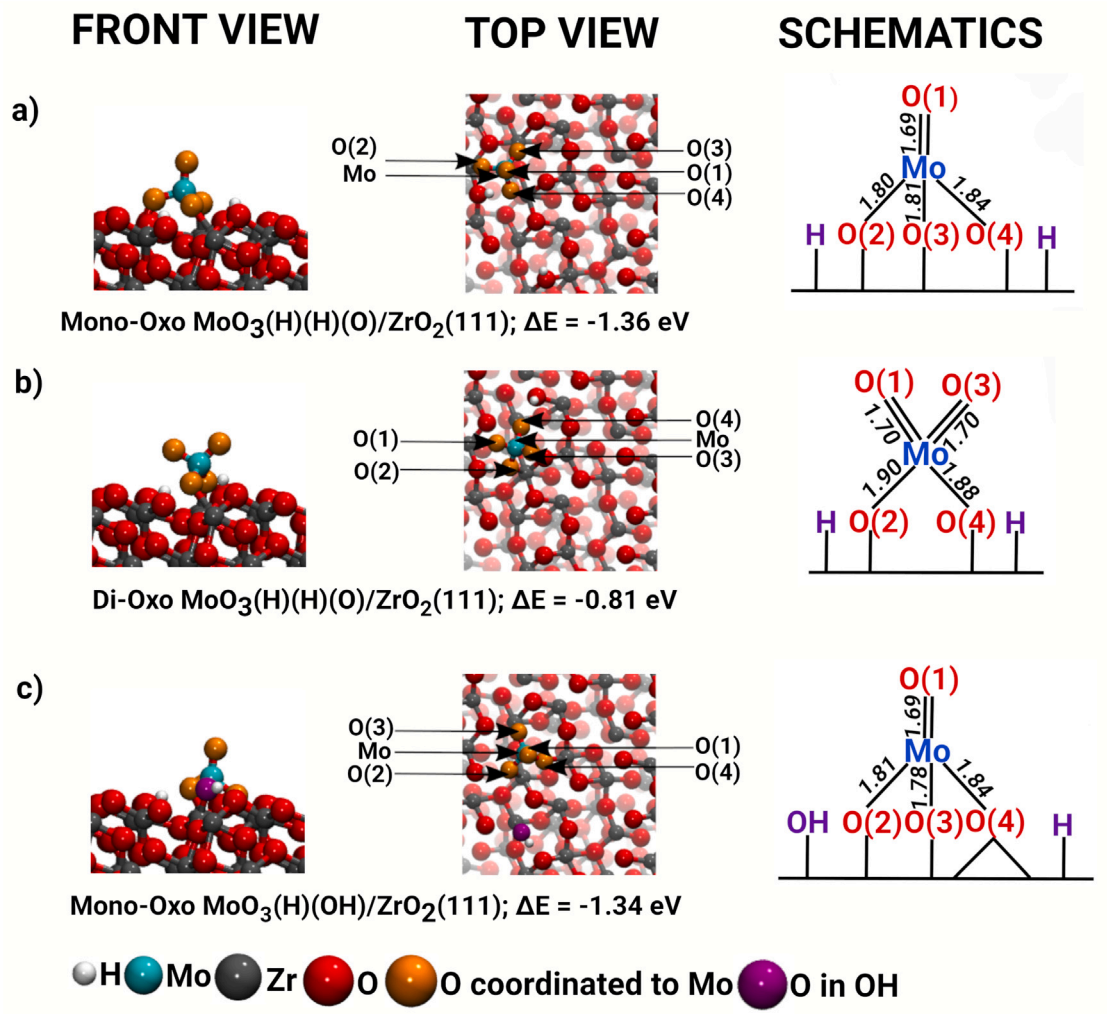

Fig. 4. Orthographic projections of molybdenum oxide moieties supported on $ZrO_2(111)$ in the presence of hydroxyls. **(a)** mono-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$, **(b)** di-oxo $MoO_3(H)(H)(O)/ZrO_2(111)$, and **(c)** mono-oxo $MoO_3(H)(OH)/ZrO_2(111)$. ΔE is the difference in energies of the products and reactants for the reaction shown in Eq. (5).

Table 2
Vibrational frequencies computed using PBE+U molybdenum oxide moieties supported on $ZrO_2(111)$ in presence of hydroxyls. The reported frequencies are scaled by a factor of 1.03.

Experimental[45]		Mono−oxo $MoO_3(H)(H)(O)$		Di−oxo $MoO_3(H)(H)(O)$		Mono-oxo $MoO_3(H)(OH)$	
Raman Freq. (cm^{-1})	Assigned	Freq. (cm^{-1})	Modes	Freq. (cm^{-1})	Modes	Freq. (cm^{-1})	Modes
970	$(O{=})_2MoO_2$ (symmetric)	973	Mo=O (stretching)	962	O=Mo=O (symmetric)	981	Mo=O (stretching)
845	Mo–O–Zr	849	O–Mo–O	942	O=Mo=O (asymmetric)	869	O–Mo–O
		771	O–Mo–O	869	O–Mo–O	801	O–Mo–O
				771	O–Mo–O	769	O–Mo–O

$$\simeq (E^{DFT}_{MW} - E^{DFT}_{M}) + (E^{vib}_{MW} - E^{vib}_{M}) - T(S^{vib}_{MW} - S^{vib}_{M}) - \mu_{H_2O} \quad (7)$$

Here E^{DFT}_{MW} is the energy of $MoO_3(H)(H)(O)/ZrO_2(111)$ [or $MoO_3(H)(OH)/ZrO_2(111)$] computed using DFT and E^{DFT}_{M} is the energy of $MoO_3/ZrO_2(111)$. E^{vib}_{MW} and E^{vib}_{M} are the vibrational contributions to energy of $MoO_3(H)(H)(O)/ZrO_2(111)$ and $MoO_3/ZrO_2(111)$; respectively. S^{vib}_{MW} and S^{vib}_{M} represents the vibrational contributions to entropy for $MoO_3(H)(H)(O)/ZrO_2(111)$ and $MoO_3/ZrO_2(111)$; respectively. Finally, μ_{H_2O} is the chemical potential of water, which we compute using the following [23]:

$$\mu_{H_2O} = \mu^{\circ}_{H_2O}(T) + RT \ln\left(\frac{P_{H_2O}}{P^{\circ}}\right). \quad (8)$$

Here $\mu^{\circ}_{H_2O}(T)$ is the reference chemical potential of water at T and P°. We plot the ΔG of the reaction (5) in Fig. 5a at different partial pressures of water at T = 623 K, which is a typical temperature for hydrodeoxygenation [5]. The effect of temperature at $P_{H_2O} = 0.03$ bar is shown in Fig. 5b. Because of inherent errors, DFT is qualitatively correct and we put little emphasis on quantitative aspects of these simulations. We find that, at low temperatures and high partial pressures of water, mono-oxo moieties with hydroxyls are energetically more favourable. The density of hydroxyls will, therefore, depend on the reaction conditions. A more important finding from these simulations is that, at all conditions, mono-oxo moieties (with or without hydroxyls)

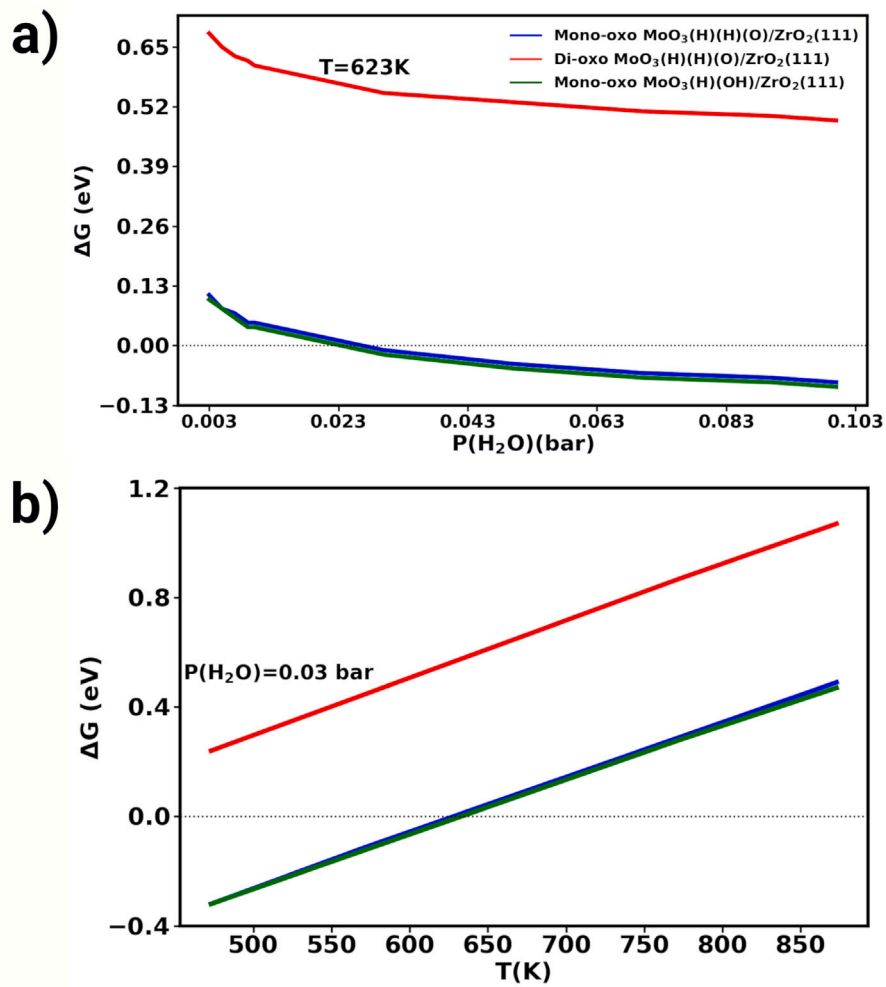

Fig. 5. Variation of Gibbs free energy of the reaction given in Eq. (5): **(a)** as a function of partial pressures of H_2O at$T$ = 623 K and **(b)** as a function of temperature at P_{H_2O}= 0.03 bar.

are more stable than di-oxo moieties; and therefore, will dominate the surface.

3.3. Dissociation of Anisole on reduced Mono-oxo Molybdenum oxide moiety

In general, it is theorized that deoxygenation of bio-oil model compounds under reducing environment of H_2 proceeds through Mars van Krevelen mechanism [5,7,84]. Wherein oxygen removal reduces Mo ions, which acts as a catalytic site for deoxygenation reactions. The oxygen vacancy is healed by an oxidant and the cycle is repeated. In a reducing environment of H_2, one can imagine the presence of a hydrogen atom adsorbates on the surface. Hydrogen is a strong base, which donates its electron to the system. This reduces Mo ion without making an oxygen vacancy. The reduced Mo ions may also be active for anisole dissociation. We investigate this below and compare the activity to reduced molybdenum oxide (bulk MoO_2) studied in our previous work [85].

We study deoxygenation of anisole to benzene, a model compound for bio-oil. Benzene has been shown to be the major product of hydrodeoxygenation using a molybdenum oxide catalyst [16,17,86,87]. The proposed pathway for hydrodeoxygenation of anisole to benzene is shown in Fig. 6a. Deoxygenation of bio-oil model compounds have been shown previously [88–90] to be an important rate-limiting step in the whole reaction network. Based on the these findings, we assume that the dissociation of anisole is one of the steps contributing significantly to the reaction rate and compute the barriers for this step. We also compute barriers for the transfer of hydrogen from the surface to adsorbed phenyl forming a gas-phase benzene molecule.

We begin our study with mono-oxo MoO_3(H)(H)(O)/ZrO_2(111), which was shown to be the most likely moiety at 600 K in the previous section. We then placed a hydrogen on the oxygen of ZrO_2(111). We tried several possible permutations and the most stable configuration was used for further exploration. The hydrogen adsorbate donates its electron to the molybdenum oxide moiety and reduces Mo from +6 to +5. We study anisole dissociation on this reduced molybdenum site. The reaction pathway is shown in Fig. 6b. We also provide a movie of the reaction steps in Supplementary Information (Figure S6). In the first step, anisole adsorbs on the Mo moiety with an adsorption energy of −0.45 eV. In the next step, proton transfers from the oxygen on ZrO_2(111) to the terminal oxygen of the molybdenum moiety. We find that the barrier for this proton migration is 1.31 eV. The next step involves the rotation of O–H, which has a low barrier of 0.3 eV. Finally, proton is transferred to the phenyl forming the benzene molecule, breaking the C–O bond. In the product state, Mo coordinates with the methoxy, returning Mo to its formal charge of +6. The deoxygenation step has a barrier of 0.93 eV, which is 0.6 eV lower than that observed by computations for anisole deoxygenation on reduced molybdenum oxide surfaces [MoO_2(100)] [85]. In comparison, the reaction where the moiety reacts with H_2 to produce H_2O and an oxygen vacancy requires an energy of 1.64 eV, which is larger than the overall barriers for anisole dissociation to benzene. Although direct

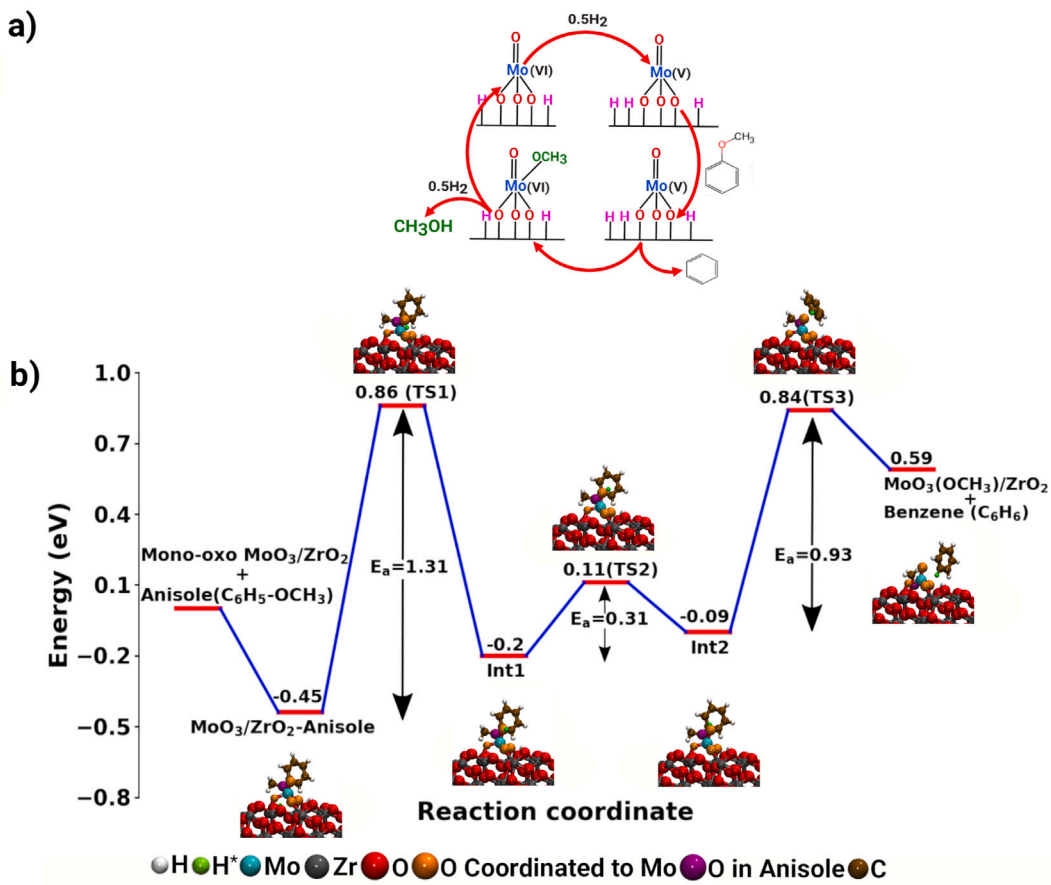

Fig. 6. **(a)** Proposed reaction pathway showing the dissociation of anisole on a MoO_3(H)(H)(O)/ZrO_2(111) in presence of H adsorbate (denoted as 'red-MoO_3'). The H adsorbate donates its electron and reduces Mo from a formal state of +6 to +5 without forming an oxygen vacancy. **(b)** Potential energy surface showing the mechanistic pathway of anisole dissociation. 'TS1', 'TS2' and 'TS3' denotes the transitions states of step 1, 2 and 3; respectively. The zero of energy is taken as non-interacting anisole and reduced molybdenum oxide moiety. The intermediates are denoted as 'Int1' and 'Int2' respectively.

comparison cannot be made, we also mention in passing that previous computational studies have reported a barrier in the range of 1.1–1.9 eV for deoxygenation of model compounds of bio-oil on an oxygen vacant molybdenum oxide surfaces [88–90]. Thus, it appears that the proposed mechanism of a hydrogen adsorbate reducing Mo ion without making an oxygen vacancy and deoxygenation of anisole on this Mo ion may be an important mechanistic process in the overall reaction network of hydrodeoxygenation of anisole.

4. Summary and conclusion

In this work, we used the DFT calculations to probe the structures and activities of molybdena moieties supported on a (111) surface of a zirconia oxide. We focused on monomeric moieties under dehydrated conditions assuming no adsorbed water is present, however, we probed the structures of molybdenum oxide in the presence of hydroxyls. Several orientations and possible structures were explored to find the most stable structures under strictly dehydrated conditions. It was found that monomeric molybdena could exists in three possible structures: (a) mono-oxo, (b) di-oxo and (c) pero-oxo configuration. The mono-oxo configuration was found to be the most stable among the three. A vibrational frequency analysis showed that the peak positions for mono-oxo configuration was more closer to that found in experiments.

Hydroxyls were created on the ZrO_2 surface by adsorbing a hydrogen on the oxygen. On adsorption, hydrogen loses it electron which gets localized on the Mo ion; thus reducing it to a lower valence state. As a result of electron localization, Mo increases its bond length with the bridging oxygens. A vibrational frequency analysis of the reduced Mo ion suggests that there is an overall red-shift in the frequencies associated with the Mo atom due to electron localization.

The electrons localized on Mo ion due to adsorption of two hydrogen atoms on ZrO_2 are strong bases and compensated by adsorbing an oxygen atom, which is a strong acid. The compensated moiety has Mo ion oxidized back to a formal state of +6. We thus obtain a monomeric molybdenum moiety with hydroxyls on the surface. Again, we find that the mono-oxo moiety is the most stable moiety and a vibrational frequency analysis gives peak positions which are an excellent match to the experimental spectra.

The stability of the monomeric molybdenum moieties were examined using ab initio thermodynamic calculations. The calculations suggests that mono-oxo moieties are the most stable moieties at all temperatures and partial pressures of water. At higher temperatures and low partial pressures of water, dehydoxylation is favoured.

The reactivity of mono-oxo moieties in the presence of a hydrogen adsorbate were examined by computing the barriers for anisole dissociation, a model bio-oil compound. The deoxygenation step to form benzene was studied on the reduced mono-oxo moiety of molybdenum. The moiety was reduced by placing a hydrogen on the support and in such conditions no oxygen vacancy is formed. It is found that the overall barriers for this step is 0.93 eV which is 0.7 eV lower than that required to make an oxygen vacancy on the moiety and also 0.6 eV lower than reported previously for anisole deoxygenation on the surface of MoO_2, which has Mo in a reduced state of +4 obtained by forming oxygen vacancies.

CRediT authorship contribution statement

Abir Lal Bose: Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Sayali Ramteke:** Data curation, Formal analysis, Investigation. **Goutam Deo:** Formal analysis, Funding acquisition, Project administration, Writing – review & editing, Validation, Supervision. **Vishal Agarwal:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jcat.2023.115267.

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